The Colombian Solar Market: Current Landscape, Challenges, and Future Outlook

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1. Executive Summary

Colombia's solar sector is on the rise, propelled by high electricity costs, favorable tax incentives, and growing consumer demand. Government policy has enabled rooftop and utility-scale expansion, yet delays in permitting, corruption, and regulatory uncertainty persist. Companies like Atlas Renewable Energy, Colombia Solar, and Emergent are leading development efforts but call for urgent reform. This report outlines Colombia's path forward, incorporating insights from project developers, consultants, and regulatory sources.

2. Introduction

Colombia is endowed with high solar irradiance (4.5–6.5 kWh/m²/day), placing it among Latin America's most promising markets for photovoltaic (PV) deployment. Yet, as of 2024, solar energy comprises only a small portion of the national electricity mix, which has long been dominated by hydropower (≈approximately 65%). Hydropower has historically provided stability but now represents a risk under intensifying El Niño patterns. The National Development Plan 2022–2026 and Colombia's revised Nationally Determined Contributions (NDCs) aim to diversify this matrix, prioritizing renewables like solar and wind to meet decarbonization targets.

3. Current State of the Solar Energy Market

Solar capacity has grown significantly, surpassing 1.5 GW in 2024, thanks to tax incentives (Law 1715/2014 and Law 2099/2021) and net metering regulations. Companies like Atlas Renewable Energy and Colombia Solar are building projects ranging from 1 to 100+ MW. According to our conversations with industry members, the internal rate of return (IRR) for solar ranges from 25% (residential) to 60% (industrial), driven by high energy costs—often exceeding $0.30/kWh.
Certain Colombian solar companies anticipate the possibility of 7 GW of rooftop solar by 2030, primarily in the commercial and industrial sectors.

4. Regulatory and Policy Framework

Colombia has a robust legislative foundation:

- Law 1715 of 2014: Tax deductions, VAT and tariff exemptions for renewables
- Law 2099 of 2021: 50% investment deduction over 15 years
- Law 2294 (PND 2022–2026): Sets energy diversification and emissions reduction goals

However, interviews with industry members highlight grid connection delays, regulatory instability, and an inactive regulatory commission (CREG) as major risks for investors.

5. Investment Landscape and Key Players

Atlas Renewable Energy

Atlas Renewable Energy, backed by Global Infrastructure Partners (GIP), launched its first Colombian solar venture—the Shangri-La PV project—with a capacity of 201 MWp, expected to generate ~403.7 GWh/year and power 214,000 homes by late 2025. The project was financed through a COP 473.8 billion (~USD 113 M) loan from IDB Invest and Bancolombia, and involves a PPA with Isagen. This investment complements Atlas's objective of achieving 1 GW of Solar capacity in Colombia.

Colombia Solar

Founded in 2015, Colombia Solar specializes in grid-tied projects ranging from 1–20 MW, primarily with foreign investors. Despite strong consumer interest and solid returns, the company faces setbacks due to bureaucratic grid-connection delays—only 5 of 220 projects were approved recently—and corruption at various government levels.

Emergent

Emergent operates across environmental consulting, solar project development, software, and e-commerce. The firm reports estimated IRRs of 25–60%, thanks to generous energy tariffs, net metering, and tech cost declines. Emergent projects 7 GW of rooftop solar by 2030, predominantly in industrial and commercial sectors.

SunColombia

Focused on distributed generation for rural and underserved regions (ZNI), SunColombia has become a leading rural solar developer. Its model includes community partnerships and NGO collaboration. While not yet widely covered in mainstream publications, the company is recognized in policy circles for its social impact approach.

Enel Green Power

The renewable division of Enel is deploying major utility-scale solar farms—such as the 486 MW Guayepo and 187 MW El Paso parks—alongside its existing wind portfolio. Enel's strategy integrates clean energy production with modern grid infrastructure to support national decarbonization targets.

Celsia

A subsidiary of Grupo Argos, Celsia is scaling urban distributed solar and pioneering energy storage. Notably, in December 2024, Celsia commissioned Colombia's first solar+storage system: a 1 MW/2 MWh $LiFePO_4$ BESS at Palmira 2 PV farm. The company also launched significant rooftop solar projects in Yumbo and Santa Rosa de Lima.

Additional Observations & Market Context

- Regulatory advancements like Law 1715 (2014), net metering (2018), and Law 2099 (2021) established tax benefits, VAT exemptions, and attractive depreciation for solar developers.
- As of early 2025, solar capacity in Colombia reached approximately 2 GW, representing about 10% of the national renewable mix—six times the level of hydropower capacities outside large hydro.

6. Challenges and Barriers

Underdeveloped Battery Policy	No clear tariff structure or market mechanism for energy storage.
Infrastructure Gaps	Lack of substations, especially in the north
Regulatory Uncertainty	No active leadership at CREG; policy reversals possible
Permitting Delays	UNE has approved only 5 of 220 projects in recent months

Corruption	Bribery required at multiple bureaucratic levels

Colombian Solar Market (2020–2025)

Colombia's solar capacity has grown rapidly under new policies and auctions. By early 2025 the country had about 2.5–3.0 GW of installed non-hydro renewables (mostly solar + wind) and expects to fall short of the 6 GW by 2026 national target. A major reliability auction in February 2024 awarded 4.4 GW of new solar projects (99% of capacity), reflecting strong developer interest. Grid-expansion (notably in Atlántico and La Guajira) and slow permitting remain bottlenecks. Nevertheless, government and regulators (UPME) continue to support large PV and distributed solar under Law 2099 (fiscal incentives, net metering, auctions).

Electric-sector players – both international Independent Power Producers (IPPs) and Colombian utilities – are rushing to build the contracted projects. The largest solar plants are concentrated in the northern Atlantic coast and interior: e.g. Enel's 187 MW La Loma (Cesar), Isagen/Atlas' 201 MW Shangri-La (Tolima), and multiple projects in Atlántico and Córdoba departments. At the same time, distributed-generation (DG) solar is expanding in

industrial and rural markets under net-metering, with local firms executing sub-MW to multi-MW rooftop and microgrid projects.

Table 1 below summarizes the major solar developers active in 2020–2025, their scale and partnerships.

Grupo Cox (Costa Rica)	0 (DG Pipeline)	15×1 MW rooftop PV plants (schools) planned (2025–26)	Part of Grupo Cox; investing COP 47B (~$11M) in distributed solar
SunColombia (Colombian DG)	<10	Numerous rural/off-grid solar-DG projects (incl. hurricane-proof microgrid)	Private firm; ~150 staff (2024); draws interest from institutional investors
EPM (Medellín utility)	83 (2024)	Tepuy PV (83 MW, Caldas); also a small wind farm (Las Pavas, 20 MW)	Municipal utility; COP 397B invested in Tepuy
Ecopetrol	0 (entering market)	Buying Statkraft's ~1.3 GW Colombia wind/solar portfolio; took 49% of AES's Jemeiwaa wind (La Guajira); negotiating Enel's wind assets	State-owned oil firm; using cash to fund renewables acquisitions
Verano Energy (Brookfield)	0 (New Projects)	Acquired 296 MW of solar: Las Palmeras (270 MW, Cesar), Chicalá (13 MW, Atlántico), Bambú (13 MW, Santander)	Subsidiary of Brookfield Renewable; ~$300 M acquisition cost

Grenergy (Spain)	90 (2023)	Inaugurated 3×30 MW (Tolima); ~1,000 MW pipeline announced (e.g. 9.9 MW Tucanes)	Equity (Spain); local debt (Bancolombia financed Tucanes)
AES Colombia (AES Corp./Chile)	>100	Solar self-gens: Castilla, San Fernando, Brisas (26 MW, Ecopetrol PPA); developing Jemeiwaa wind (1.1 GW, La Guajira)	Owned by AES Andes; finance via corporate PPAs (e.g. with Ecopetrol); target 2 GW total by 2030
Atlas Renewable Energy (GIP, USA)	0 (pipeline)	Shangri-La PV – 201 MWp (160 MW_AC) in Tolima (COD late-2025); target 1,000 MW over ~10 yrs	Backed by Global Infrastructure Partners (GIP); IDB Invest + Bancolombia loan (COP 473b, ~$113m); PPA with Isagen
Cubico Sustainable (UK)	300	200 MW under construction; ~500 MW late-stage dev (with Celsia)	JV with Celsia (49/51); backed by Cubico's global renewables funds
Celsia (Grupo Agos, Colombia)	~300	6 PV farms (Valle/Tolima, totalling 59.4 MW); 79.5 MW under dev + 375 MW acquired from Mainstream; 1 MW/2 MWh BESS (Palmira)	COP 540B planned investment (2024); joint-venture with Cubico (49%); Argos equity
Enel Colombia (Enel S.p.A., Italy)	~688	La Loma (187 MW), El Paso, Fundación; Guayepo I&II (2×243 MW); **1.2 GW** won in 2024 auction	€300M loan (EIB/SACE) for Guayepo; parent Enel ownership; AAA rating

Each developer shows strategic positioning. For example:

- Enel Colombia (subsidiary of Italy's Enel) is the largest renewables generator (≳4.0 GW total), with 688 MW solar (≈18.7% of Colombia's solar). It has strong finances (Fitch AAA) and major projects in hand. In 2024 Enel completed its 243 MW

Guayepo I & II PV parks (with an EIB/SACE-backed US$300m loan) and broke ground on Guayepo III and Atlántico (~700 MW total). In the Feb. 2024 auction Enel won ~1.2 GW, underlining its market leadership. Strengths: deep pockets, integration with a 25%-market-share distribution business, global expertise. Weaknesses: project execution relies on grid expansion and permitting; currency exposure.

- Celsia (Grupo Argos) is Colombia's largest domestic power company (Valle del Cauca/Tolima grid owner) and second-ranked renewables developer. By early 2024 Celsia had ~300 MW of PV operating. It built six PV plants (Yumbo, Palmira, etc.) totalling 59.4 MW, and has ~79.5 MW more under construction plus a 375 MW bulk portfolio acquired from Mainstream Renewable Power in 2025. Celsia also installed a 1 MW/2 MWh battery at Palmira 2 – reportedly Colombia's first PV+storage farm. Argos Group has earmarked ~COP 540 billion to boost Celsia's renewables. Celsia's strengths are its established local utility network, corporate balance sheet, and a new JV with UK fund Cubico (to co-invest). Its weaknesses include limited track record in development (initial projects were <20 MW) and dependence on partner Cubico and Argos funding for scaling up.

- Cubico Sustainable Investments (UK) partnered with Celsia to accelerate scale. The Cubico–Celsia JV operates ~300 MW of PV in Colombia (commissioning 200 MW in 2023) and has ~700 MW more in the pipeline. It also holds Celsia's legacy transmission & distribution assets (Caoba T&D). Cubico's global funds bring capital and expertise, but as a minority partner, its influence is aligned with Celsia's strategy.

- Atlas Renewable Energy (backed by GIP, USA) entered Colombia in 2024. Its first project is Shangri-La, a 201 MWp (160 MW AC) solar farm in Tolima (COD late 2025), under a 15-year PPA. Atlas plans to build up to 1 GW in Colombia (in partnership with public gen-co Isagen). Globally, Atlas has built ~5 GW (2.7 GW operational). Its advantages include institutional backing (GIP/IDB) and PPA contracts; its risks are reliance on favorable regulation and competition from incumbents. Atlas developed the Shangri-La project, contributing 403 GWh/year—enough to power 214,000 homes. Its success relied on long-term foreign financing and partnerships with local utility Isagen. The company plans to develop 1,000 MW of solar in Colombia.

- AES Colombia (part of AES Corp.) focuses on self-generation PPAs. It has >100 MW of solar assets (Castilla, San Fernando, 26 MW Brisas selling to Ecopetrol), and is now building wind (the 1.1 GW Jemeiwaa in La Guajira). AES's strength is technical experience and corporate contracts; its weakness is a focus on the captive market and less retail presence.

- Grenergy (Spain) is developing utility PV at mid-scale. By 2023 it commissioned ~90 MW (three 30 MW plants in Tolima) and has an initial Colombian pipeline of ~1,000 MW. Its advantage is agility and global backlog (5 GW in LatAm). A challenge is competition from deeper-pocketed IPPs and dependence on local finance (e.g. Bancolombia loan).

- Verano Energy (a Brookfield Renewables arm) is an up-and-coming entrant. In 2023, it acquired three solar park projects (270 MW Las Palmeras plus two 13 MW sites) for ~$300M. Verano, experienced in Chile and Peru, has ~3 GW total LatAm portfolio and is transitioning from developer to owner-operator. Its strengths are project pipeline and Brookfield support; risks include the need to actually build-out the acquired projects by 2026.

- Ecopetrol (state oil company) is now a major consolidator in renewables. In 2025, it agreed to buy Statkraft/Enerfin's entire Colombian solar+wind portfolio (~1.3 GW). It also took 49% of AES's Jemeiwaa wind and is negotiating Enel's wind projects. Ecopetrol's deep pockets and local clout are strengths, but it has limited experience in renewables.

- EPM (Empresas Públicas de Medellín) is a local public utility. In 2024, it completed its first utility solar project, the 83 MW Tepuy farm (Caldas) (COP 397B invested). EPM also has a 20 MW wind farm (Las Pavas). As a stable municipal company, EPM is financially secure, but historically focused on hydro/gas and only now ramping up solar.

Distributed Solar Companies

Beyond big IPPs, Colombian solar has many distributed-generation (DG) providers serving industry, institutions, and rural areas. These tend to be smaller Colombian or regional firms, often taking advantage of net-metering and solarization programs:

- SunColombia SAS – Founded in 2012, this Bogotá-based firm specializes in solar hybrid systems for off-grid communities, industries, and emergency relief. It has installed numerous microgrids and "pump-as-turbine" solar-water projects (e.g. post-Hurricane Providencia). It emphasizes innovation and social impact, touting ~150 employees (2024) and expansion plans to other Caribbean countries. Its strength is local knowledge and technical patents; its weakness is small scale and dependence on subsidies/investors.

- Grupo Cox – A Costa Rican utility venture. In mid-2025 it announced 15 new 1 MW rooftop solar projects (15 MW total) for Colombian schools (fully privately financed by

Grupo Cox). This highlights foreign interest in Colombia's DG segment.

- Other local installers and ESCOs – Dozens of Colombian SMEs and startups (e.g. Energías Renovables Bogotá, Grupo Zener, etc.) are active in commercial rooftop, rural mini-grid, and residential solar. Many of these have been slow to scale due to bureaucratic and financing hurdles; Law 2099 and public credit programs (Bancóldex, IDB) aim to grow this segment.

Strategic Analysis: Strengths & Weaknesses

Each company's position reflects a mix of strategic strengths and weaknesses in the Colombian context:

- Financing & Partnerships: International players (Enel, Atlas, Grenergy, Cubico/Argos) generally have strong access to capital and global expertise, often partnering with development banks (EIB for Enel, IDB for Atlas) or local institutions (Bancolombia loans). State and municipal firms (EPM, Ecopetrol) rely on government funding. Smaller developers and DG firms depend on equity or project financing (e.g. Bancóldex loans to Celsia/Cubico, private equity for SunColombia).

- Regulatory Alignment: Most major players aligned with the national auction and policy framework: they won OEF contracts in 2024 or hold government-backed PPAs. For example, Atlas's Shangri-La was submitted to the reliability auction and won. However, policy shifts or permitting bottlenecks can be a weakness: e.g., EDF (French utility) abandoned a 43 MW plant in 2024 over land/permits (though that source is anecdotal).

- Technical Capability and Scale: Enel, AES, and to a lesser extent Cubico/Atlas have large technical teams and experience with multi-hundred-MW projects. Celsia and EPM are utilities with some engineering capacity. Smaller DG firms (SunColombia, Grupo Cox) excel in niche tech (solar+diesel, microgrids) and social projects but cannot build GW-scale farms.

- Local Engagement: Domestic companies (Ecopetrol, Argos/Celsia, EPM) benefit from local political support and knowledge. International entrants often create local partnerships (e.g., Atlas with Isagen, Cubico with Celsia). Community relations are critical in projects (Colombia requires "Consulta Previa" with indigenous groups), and firms like Atlas and Enel emphasize social programs. Weakness: foreign firms must adapt to local procurement, labor, and land-use rules; domestic firms must attract global capital.

- Innovation: Some companies stand out for innovation: for example, Celsia's hybrid PV+storage and SunColombia's solar-wind-diesel hybrids. Overall, the market is less

about cutting-edge tech and more about finance/scale, but emerging trends (batteries, hydrogen) will be areas to watch.

7. Competitive Landscape & Collaborations

The landscape is competitive but also collaborative:

- Auctions: The 2024 Reliability Charge auction saw Enel win ~25% of bids (1.2 GW). Other winners include Verano (270 MW Las Palmeras) and domestic utilities. With 30 projects awarded, developers now compete on building capacity by 2027–28.

- Joint Ventures and M&A: Key partnerships include the Celsia–Cubico JV, which pools local and foreign capital. Atlas's entry via M&A (acquiring Shangri-La from Rayo/Black Orchid) exemplifies growth via acquisition. Ecopetrol's purchases of Statkraft and AES assets show consolidation. Developers often use project-level SPVs, but investors are increasingly forming utilities-style portfolios (e.g., Argos/Celsia).

- Corporate Off-takers: Several projects are self-generated for large companies (Colombian oil majors, cement firms, palm oil, etc.). For instance, AES's Brisas solar has a long-term PPA with Ecopetrol. These corporate deals reduce market risk but tie revenues to a few customers.

- Supply Chain: Colombia imports most PV panels and inverters. Some assembly (trucks, racking) is local. Policy encourages local content but enforcement is nascent. No major module manufacturer is based in Colombia yet. Developers compete on finance and land rather than equipment.

8. Future Trajectory

Growth Outlook: Given current plans and auctions, Colombia could reach ~2.6 GW NCRE by the end of 2025 and ~3 GW by 2026, still under the 6 GW goal. Many projects (especially large ones) won't be operational until 2027–28. To bridge the looming supply gap (2–6 GW by 2028), policymakers may accelerate permits or hold new auctions.

Technology Trends: More solar+storage hybrids are expected (Celsia's Palmira farm was a pioneer). Off-grid and microgrid solutions will expand in rural zones (DG firms like SunColombia will play a role). Green hydrogen is on the horizon (Colombia passed a

Hydrogen roadmap); future co-located PV could feed electrolyzers, though this is early-stage.

Market Structure: The winners of upcoming projects will likely be the well-capitalized IPPs and utilities. Consolidation may continue (Ecopetrol's moves indicate an industrial megaproject strategy). Retail electricity prices (and demand) will shape investment; current wheeling rules and VRE integration regulations will be critical.

Local vs. Foreign Balance: Colombian firms with strong balance sheets (Argos/Celsia, Ecopetrol, EPM) are solidifying their market share. International players (Enel, AES, Cubico, Atlas, Grenergy, Verano) will supply the financing and know-how needed for gigawatt-scale builds. Collaboration between these groups – through JVs, consortia or market mechanisms – will determine how quickly the country can meet its renewable targets.

In summary, Colombia's solar market (2020–2025) is driven by a mix of global IPPs and savvy local companies, backed by international banks and development funds. Strategic strengths include strong pipelines, access to auctions, and ample sunlight. Challenges are regulatory delays and competition for land/Grid. Going forward, the industry is poised for rapid expansion and increasingly sophisticated projects, from utility-scale parks down to community solar – provided that policy support and investment financing keep pace.

9. Distributed Generation and Electrification of Remote Areas

Distributed generation (DG) is increasingly seen as a solution for regions like La Guajira and the Amazon, where grid access is poor. Emergent and Colombia Solar both report a shift toward rooftop and off-grid systems, encouraged by:

- Net metering (2018)
- Accelerated depreciation
- High returns on self-consumption projects

Battery storage is becoming critical in unreliable grid zones but still lacks the legal and market clarity needed for scale.

10. Environmental and Social Considerations

Permitting processes must comply with consulta previa regulations in territories inhabited by indigenous and Afro-descendant communities. Projects have been delayed due to a lack of stakeholder engagement and perceived environmental risks.

New regulatory updates are also boosting run-of-river hydro projects, expanding the scope of tax-exempt renewable investments beyond solar.

11. Recommendations

1. Appoint full leadership at CREG and issue clear regulatory guidelines for energy storage.
2. Streamline project approval timelines via centralized digital platforms.
3. Expand transmission infrastructure, particularly in the north.
4. Develop financial products to unlock residential solar investment.
5. Combat corruption through process transparency and third-party audits.
6. Incentivize storage systems with hourly tariffs and financing mechanisms.